Jingwen (Lisa) Zhong
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EDUCATION

University of Toronto **Toronto, ON**
Bachelor of Science in Data Science Specialist and Actuarial Science Major Expected Graduation May 2026

- **GPA: 3.94/4.0; Major GPA: 3.97/4.0**
- **Relevant Courses**: Machine Learning, Data Structure & Algo Design, Time Series, Probability, Data Science App I&II
- **Awards**: University of Toronto Excellence Award (UTEA) for 2024 & 2025 ($15,000), Department of Computer Science Research Award 2025 ($7,500 declined), The Robert Bruce Scholarship and Bursary Fund ($1,000), Dean's List for all Academic Years, New College Council In-Course Scholarship ($750), Actuarial Science Conference Travel Award ($200)
- **Teaching Assistantships**: STA130 (Intro to Stat Reasoning), MAT223 (Linear Algebra I), MAT224 (Linear Algebra II)

PUBLICATIONS

Peer-Reviewed Journal

Wortmann, U. G., Tsan, T., Niazi, M., A, I., MA, Navasardyan, R., Marun, M., Chede, B. S., **Zhong, J.**, & Wolfe, M. (2025). The Earth Science Box Modeling Toolkit (ESBMTK 0.14.0.11): a Python library for research and teaching. Geoscientific Model Development, 18(4), 1155–1167. https://doi.org/10.5194/gmd-18-1155-2025

Preprint (preparing submission to JMLR (Journal of Machine Learning Research))

Xiang, Y., **Zhong, J.**, Yan, Y., Koutrakis, P., Garshick, E., & Franklin, M. (2025). Spatiotemporal Satellite Image Downscaling with Transfer Encoders and Autoregressive Generative Models. https://doi.org/10.48550/arXiv.2512.05139

RESEARCH EXPERIENCES

The Modeling and Animation of Extreme Weather Events – Supervisor: Dr. Ron Baecker Aug 2025 – Present
Department of Computer Science & Dynamic Graphics Project, University of Toronto

- Building an end-to-end **inference pipeline** for wildfire forecasting using Microsoft's Aurora model, producing fire probability by deriving the Fire Weather Index (FWI) and injecting it into the Fire Behavior Prediction (FBP) system.
- Implementing scalable **Python** workflows on Linux clusters for Aurora inference and downstream wildfire modeling, ensuring efficient data processing and evaluation of fire-danger predictions.
- Producing geospatial visualizations of predicted fire hazard rates using ggplot and ggspatial in **R**, enabling landscape-level communication of climate-driven wildfire risk.

Satellite Image Downscaling – Supervisor: Dr. Meredith Franklin May 2025 – Present
Department of Statistical Sciences, University of Toronto

- Developed **deep learning** and **transfer learning** models for resolution calibration and downscaling using satellite data, improving spatial resolution from 50 km to 7 km and achieving a high predictive performance of mean $R^2 \approx 0.9$.
- Pretrained on a **UNet** model to learn spatial-temporal dynamics and transfer to a large DDPM model, using **time series** dust extinction data, static covariates (elevation, latitude/longitude) to generate high-resolution dust extinction data.
- Enhanced accuracy further through region- and season-specific training with cross-validation and early stopping, reaching R^2 up to 0.94 and RMSE < 0.02.
- Co-authoring a manuscript detailing our downscaling framework for submission to a peer-reviewed journal.

LLM in Patent Analysis – Supervisor: Dr. Loren Brandt & Dr. Kevin Lim Sep 2024 – Sep 2025
Department of Economics, University of Toronto

- Implemented a refined **unsupervised topic modeling** pipeline to classify over 260,000 UAV patent dataset using Cohere embeddings, UMAP, HDBSCAN, and BERTopic, achieving a 30% reduction in output topics.
- Enhanced pipeline robustness and accuracy through hyperparameter **finetuning** and embedding comparisons between abstracts and claims, and integrated the OpenAI API into the pipeline, resulting in improved clustering performance.
- Mapped patents into value-chain and patentee taxonomies with keyword extraction, semantic distance, and entity resolution to investigate industry-specific innovation patterns across firms, universities, and individuals.
- Provided visual analytics results, including topic maps and similarity heatmaps to highlight research findings.

AI and Machine Learning in Medical Imaging – Supervisor: Dr. Pascal Tyrrell May 2024 – Sep 2024
Prof. Tyrrell's Lab (Report, Poster)

- Benchmarked image quality metrics, including SSIM, PSNR, and LPIPS, on GANs and diffusion model outputs for medical images under degradations like image compression, noise signals, and downsizing using **Python**.
- Optimized tensor operations on the GPU level using **TensorFlow** and **PyTorch**, efficiently processed large-scale datasets.
- Reconciled and validated research findings with ANOVA test in **SAS**, achieved statistical significance of p-value < 0.001, and concluded with optimal performance provided by LPIPS.

Oceans' long-term response to IPCC emission scenarios – Supervisor: Dr. Ulrich Wortmann May 2024 – Aug 2024
Department of Earth Sciences, University of Toronto (Poster)

- Delivered a model to predict long-term impacts on Carbon Dioxide (CO_2) emissions on the ocean system and back tested using historical climate data.
- Optimized the hypsometric modeling code using the grid method and multi-threading to improve efficiency by 110%.
- Developed visualized results using **Matplotlib** for long-term climate impacts comparison under the RCP scenarios.

- Co-author and contributing developer of ESBMTK, a published open-source climate modeling toolkit adopted in academic research and teaching (published in Geoscientific Model Development 2025).

WORK EXPERIENCES

Actuarial Analyst Intern
Aviva Canada

Actuarial and Data Science Intern Jan 2026 – Present

- Designed and automated a loss data ingestion and reconciliation pipeline using **SAS, SQL, and Python**, standardizing datasets across cross-functional teams and reducing manual validation effort
- Supported a major pricing model refresh filing by reconciling multi-source actuarial outputs in Excel, ensuring audit-ready documentation and regulatory compliance

Actuarial Analyst Intern, Ontario Auto Street Pricing May 2025 – Aug 2025

- Supported actuarial pricing by leveraging **SAS**, **SQL,** and **Earnix** to build analytical tools and validate models, ensuring accurate and robust pricing strategies.
- Pre-processed and analyzed loss and exposure data through automated scripts in **Excel VBA** and **Python**, reducing data preparation time by 30% and streamlining pricing reviews**.**
- Improved model accuracy and consistency by reconciling discrepancies across data sources and eliminated 98% of data mismatches between variables and premium values.
- Designed and delivered broker-facing presentations on pricing changes, dislocation, and customer impact, effectively clarifying the rationale behind segmentation updates.

Data Analytics Intern May 2024 – Sep 2024
Temerty Faculty of Medicine, University of Toronto

- Developed and deployed an integrated full-stack information management system with **MySQL** and **Power BI** integration using **JavaScript**. Provided a full-cycle data collection, analysis, and visualization service from user to admin.
- Built an end-to-end **data pipeline** for the collection and analysis of data for visualization for the budgeting department.
- Processed and analyzed data using **Excel VBA** and developed interactive dashboards in **Power BI** to visualize KPIs of budget variance, departmental expenditures, and forecasting accuracy, supporting evidence-based decision-making.

PROJECT

CAS Case Competition

Hurricane Insurance Visualization (First Place). Feb 2025

- Developed an interactive **full-stack** web application using **JavaScript**, D3.js, and Leaflet.js, implemented a front-to-end architecture that allows users to filter historic hurricanes by different keywords from **PostgreSQL** database.
- Pre-processed and structured raw storm datasets from CSV files using **Python** scripts and **SQL** for database construction.
- Implemented a data pipeline that fetches data from SQL database by constructing a proper query based on the request.
- Awarded as the regional champion for the 2025 CAS Case Competition.

Home Insurance Pricing (North America Regional Champion) Feb 2024

- Developed a GLM-based discount strategy for house insurance by analyzing consumer data that balanced profitability and market competitiveness using **Python**.
- Performed segmentation analysis to model market response toward discount activities under adverse options.
- Awarded as the North America regional champion for the 2024 CAS Case Competition.

Learning Agency Lab – PII Detection (Kaggle Bronze Medal, Top 9%) Feb 2024 – May 2024

- Developed and fine-tuned an **LLM** for personal identifiable information detection using DeNERTa-v3 on 7,000 essays.
- Augmented training dataset with 11,000+ essays generated via OpenAI API and automated semantic entity extraction, improving model robustness.
- Achieved **95.74% F1 score**, earning a **bronze medal** and placing in the **top 9%** out of 2,050 teams in Kaggle's PII Detection & Removal from Educational Data competition.

LEADERSHIP

VP Career&Academic @ UTChinese Network May 2025 – Present

- Leading team of 30 to organize career fairs, panel talks, and firm visits in various industries for the community.

Academic Development Outreach Director @ Students in Data Science and Statistics (SDSS) May 2025 – Present

- Reaching out to industry/academic professionals and planning data science-related academic events.

Project Manager @ ACE Career Fair Sep 2024 – May 2025

- Managed a team of 15 to hold a career fair for over 200+ students and 7 industrial partners. Oversaw logistics, sponsorship communications, and industry outreach, achieving a 40% growth in project impact.

Event Director @University of Toronto Machine Intelligence Student Team Sep 2024 – May 2025

- Conducted events for the AI/ML-related events featuring relevant industry speakers for community enrichment.

SKILLS

- Programming Languages: Python (NumPy, Pandas, Matplotlib, Scikit-Learn), R, SQL, Java, JavaScript, C
- Technical Proficiencies: PyTorch, TensorFlow, Tableau, Power BI, Earnix